

Micro Set 5. Public Goods and Information Failure — National Defence and Used-Car Markets

Instructions. This booklet contains a single case study on public goods and information failure, followed by six short-answer questions ranging from 2 to 6 marks.

Section Case Study. Two Failures of the Free Market: National Defence and the Used-Car Market

Table 1.1: Selected Public Goods and Information Failure Indicators, 2019–2024

Indicator	2019	2020	2021	2022	2023	2024
Singapore Defence Spending (S\$ billion)	15.5	15.0	15.5	16.4	17.0	19.8
Defence Spending (% of GDP)	3.0	3.2	2.8	2.8	2.7	2.9
National Service Active Strength (thousands)	71	71	71	71	70	70
National Library, public visits (millions)	17.2	11.5	12.6	15.1	17.6	18.4
Used Cars Sold (units)	79100	71200	73000	65800	60200	54400
Used-Car Lemon-Law Complaints (units)	1840	1620	1980	2350	2680	2520
Mean Used-Car Price (S\$ thousands)	96.0	92.0	110.0	142.0	138.0	125.0

Source: Adapted from Ministry of Defence Singapore, National Library Board, Land Transport Authority, and Consumers Association of Singapore (CASE).

Extract 1.1: Two failures of the free market: defence and the used-car lemon problem

Pure public goods: national defence. National defence is the textbook example of a pure public good. It is *non-excludable*: once Singapore’s airspace is protected by the RSAF and its territorial waters by the RSN, no citizen or resident within Singapore can be excluded from the protection. And it is *non-rivalrous*: one resident’s enjoyment of protection does not diminish another’s. These two properties give rise to the free-rider problem: in a hypothetical free market, every potential beneficiary would have an incentive to wait for others to pay for defence while consuming the protection that results. The aggregate willingness to contribute would be far below the socially optimal level. “This is why national defence is funded universally through taxation and provided by the state,” the Ministry of Defence states. “It cannot work otherwise.” Singapore’s defence spending was S\$19.8 billion in 2024, equivalent to approximately 2.9 per cent of GDP, with the National Service framework ensuring an active strength of around 70000 servicemen.

The National Library system, while not a pure public good (a library book is rivalrous — only one reader can borrow it at a time), exhibits sufficient non-rivalry in many of its services (digital archives, reading rooms, learning programmes) to merit public provision under similar logic. Public visits rose from 17.2 million in 2019 to 18.4 million in 2024.

Information failure: the used-car market. The market for used cars in Singapore illustrates a different form of market failure — asymmetric information. Sellers (typically dealers or owners selling on consignment) know far more about the vehicle’s true condition, accident history, and likely future maintenance costs than potential buyers. As described by George Akerlof in his 1970 paper “The Market for Lemons,” this information asymmetry causes the market to fail in a specific way: buyers, unable to verify quality, are willing to pay only an average price reflecting

the average expected quality. Sellers of genuinely high-quality cars receive below their reservation price and withdraw, leaving the market dominated by low-quality cars (“lemons”).

Singapore’s used-car market reflects this dynamic. Lemon-law complaints to the Consumers Association of Singapore (CASE) rose from 1840 in 2019 to a peak of 2680 in 2023, even as the volume of transactions fell from 79100 to 54400 over the same period — meaning the complaint rate per transaction rose more sharply still. The Lemon Law (introduced in 2012 under the Consumer Protection (Fair Trading) Act) requires dealers to repair, replace, or refund vehicles found defective within six months of sale. “The law has reduced but not eliminated the lemon problem,” a CASE spokesperson told the Business Times. “Dealers can still misrepresent vehicle history at the point of sale, particularly accident damage and odometer readings.”

The Land Transport Authority’s One Motoring portal now provides centralised access to vehicle inspection records, accident history (subject to data-protection rules), and registration details. This is a textbook intervention to correct information asymmetry: rather than regulating the transaction, it equips buyers with the information needed to make informed decisions. “Disclosure is more efficient than regulation when the underlying problem is information,” a market-design economist at NUS observed.

Source: Adapted from MINDEF Annual Report 2024, CASE Annual Reports 2019–2024, and Akerlof, G. (1970), “The Market for Lemons,” Quarterly Journal of Economics.

Question 1**[2 marks]**

Define a “pure public good” and identify two characteristics from Extract 1.1 that make national defence one.

Question 2**[2 marks]**

With reference to Table 1.1, describe the trend in used-car lemon-law complaints between 2019 and 2024.

Question 3**[4 marks]**

Using evidence from Extract 1.1, explain why the free market would fail to provide an adequate quantity of national defence.

Question 4**[4 marks]**

Explain, with reference to Extract 1.1, how asymmetric information in the used-car market generates the “market for lemons” problem.

Question 5**[6 marks]**

With the aid of a diagram, explain why the free market under-allocates resources to pure public goods such as national defence.

Question 6**[6 marks]**

Using evidence from Extract 1.1, explain why centralised information disclosure (e.g. the One Motoring portal) may be a more efficient policy response to the used-car lemon problem than direct regulation.

End of Set 5 — Questions